

Typo and Query Modification Strategies

Random Insertion

Description: Randomly inserts characters into a word.

Implementation: Selects a random position in the word and inserts a random letter.

Example:

Original: Where is the capital of Germany?

Modified: Where is thea capital of Germany?

Random Deletion

Description: Randomly deletes characters from a word.

Implementation: Selects a random position in the word and removes the character at that position.

Example:

Original: Where is the capital of Germany?

Modified: Where is th capital of Germany?

Random Replacement

Description: Randomly replaces characters in a word with adjacent keys on the keyboard.

Implementation: Selects a random character in the word and replaces it with a random adjacent key on the keyboard.

Example:

Original: Where is the capital of Germany?

Modified: Where is the capitsl of Germany?

Random Repetition

Description: Randomly repeats characters in a word.

Implementation: Selects a random position in the word and duplicates the character at that position.

Example:

Original: Where is the capital of Germany?

Modified: Where is the capittal of Germany?

Random Swapping

Description: Randomly swaps adjacent characters in a word.

Implementation: Selects a random position in the word and swaps the character with its adjacent character.

Example:

Original: Where is the capital of Germany?

Modified: Where is hte capital of Germany?

Common Misspelling Words (CMW)

Description: Replaces words with their common misspelled variants.

Implementation: Checks if a word is in a predefined dictionary of common misspellings and replaces it if found.

Example:

Original: Where is the capital of Germany?

Modified: Where is the capitol of Germany?

Random Letter Case Change

Description: Randomly changes the case of letters in a word.

Implementation: Selects a random letter in the word and changes its case (upper to lower or vice versa).

Example:

Original: Where is the capital of Germany?

Modified: Where is the caPital of Germany?

Synonym Replacement

Description: Replaces words with their synonyms.

Implementation: Uses NLTK's WordNet to find synonyms for a randomly selected word and replaces it.

Example:

Original: Where is the capital of Germany?

Modified: Where is the metropolis of Germany?

Add Noise Characters

Description: Adds random noise characters into words.

Implementation: Inserts random punctuation or digit characters into a randomly selected word.

Example:

Original: Where is the capital of Germany?

Modified: Where is the cap1ital of Germany?

Add Taxonomy

Description: Adds a confusing taxonomy choice at the start of the query.

Implementation: Prepends a random item from a predefined taxonomy list to the query.

Example:

Original: Where is the capital of Germany?

Modified: Munich. Where is the capital of Germany?

Repeat Key Words

Description: Repeats key words in the questions.

Implementation: Selects a random word in the query and duplicates it at a random position.

Example:

Original: Where is the capital of Germany?

Modified: Where is the capital capital of Germany?

Language Switching

Description: Inserts words from a foreign language.

Implementation: Inserts a random foreign word from a predefined list into the query.

Example:

Original: Where is the capital of Germany?

Modified: Where is the hola capital of Germany?

Random Character Substitution

Description: Substitutes characters with visually similar numbers or symbols.

Implementation: Replaces certain letters with visually similar numbers or symbols (e.g., 'O' with '0').

Example:

Original: Where is the capital of Germany?

Modified: Where is the capit@l of Germany?

Random Foreign Word Insertion

Description: Inserts random foreign words into the word list.

Implementation: Inserts a random foreign word from a predefined list into the query.

Example:

Original: Where is the capital of Germany?

Modified: Where is the bonjour capital of Germany?

Random Phrase Translation

Description: Translates random words or phrases to a random foreign language and back.

Implementation: Uses a translation API to translate a random word to a foreign language and back.

Example:

Original: Where is the capital of Germany?

Modified: Where is the Hauptstadt of Germany?

Language-based Structural Transformation

Description: Transforms sentence structure to reflect foreign language syntax.

Implementation: Rearranges the word order based on the syntax of a randomly chosen foreign language.

Example:

Original: Where is the capital of Germany?

Modified: Where Germany of the capital is?

Emoji Substitution

Description: Replace words with related emojis or insert emojis into the query.

Implementation: 1. Create a dictionary mapping words to related emojis. 2. Randomly select words in the query. 3. Replace selected words with their emoji equivalents or insert emojis after them.

Example:

Original: What is the weather like today?

Modified: What is the 🌤️ like today? 🌤️

Text-to-Speech Mishearing Simulation

Description: Modify words to simulate common speech recognition errors.

Implementation: 1. Create a list of common homophones and near-homophones. 2. Scan the query for words that have homophones. 3. Randomly replace words with their homophones.

Example:

Original: How to pair my Bluetooth device?

Modified: How to pear my Bluetooth device?

Autocorrect Gone Wrong

Description: Simulate autocorrect mistakes by replacing words with similarly spelled but incorrect words.

Implementation: 1. Create a dictionary of common autocorrect mistakes. 2. Scan the query for words that match keys in the dictionary. 3. Replace matching words with their incorrect 'autocorrected' versions.

Example:

Original: How to make duck confit?

Modified: How to make duck confident?

Internet Slang Insertion

Description: Insert or replace words with common internet slang and abbreviations.

Implementation: 1. Create a dictionary of internet slang terms and their meanings. 2. Randomly select words in the query. 3. Replace selected words with their slang equivalents or insert slang terms.

Example:

Original: What are the best movies to watch?

Modified: What are the best movies to watch? IMHO TBH

Punctuation and Capitalization Removal

Description: Remove punctuation and capitalization to mimic quick, informal typing.

Implementation: 1. Remove all punctuation marks from the query. 2. Convert the entire query to lowercase.

Example:

Original: What's the capital of France? Is it Paris?

Modified: whats the capital of france is it paris

Keyword-Only Query

Description: Reduce the query to essential keywords, simulating users who type minimal search terms.

Implementation: 1. Identify stop words (common words like 'the', 'is', 'are'). 2. Remove stop words and retain only key terms.

Example:

Original: What are the symptoms of the common cold?

Modified: symptoms common cold

Symbolic Substitution

Description: Replace words or parts of words with symbols or emojis that visually resemble them.

Implementation: 1. Create a dictionary mapping letters or words to visually similar symbols or emojis. 2. Scan the query for matches in the dictionary. 3. Replace matches with their symbolic or emoji counterparts.

Example:

Original: How to solve for x in algebra?

Modified: H👉w t👉 s👉lve f👉r X in ∇lgebra? 🍷